

# TinySea Simulation Variable Reference

Complete reference for every configurable parameter, runtime variable, constant, and output field in the simulation system. Cross-referenced with source code as of the current [simulation](#) branch.

## SimulationConfig

Source: [Assets/scripts/Simulation/DataStructure/SimulationConfig.cs](#)

Type: ScriptableObject (editable in Unity Inspector)

### Simulation Timing

| Field                             | Type | Default | Range    | Description                                                                                   |
|-----------------------------------|------|---------|----------|-----------------------------------------------------------------------------------------------|
| <a href="#">BiologyStep</a>       | int  | 1       | 1-5      | Days between biology calculations. 1 = daily (most accurate), 5 = original game behavior.     |
| <a href="#">DaysPerScenario</a>   | int  | 365     | 1-182500 | Number of days per scenario. 35 = quick test, 365 = 1 year, 3650 = 10 years.                  |
| <a href="#">NumberOfScenarios</a> | int  | 5       | 1-100    | How many times to run the scenario with different random seeds. Enables statistical analysis. |

### Carrying Capacity

| Field                                 | Type  | Default | Range                                                                                                                                                                                                         | Description                                                                                         |
|---------------------------------------|-------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| <a href="#">UseCarryingCapacity</a>   | bool  | true    | Enable soft population limit for Tier 1. Reduces birth rate as population approaches the limit. Does NOT kill creatures. Formula: $\text{births} = \text{rawBirths} * (1 - \text{tierPop}/\text{capacity})$ . |                                                                                                     |
| <a href="#">CarryingCapacityTier1</a> | float | 5000    | 100-100000                                                                                                                                                                                                    | Maximum sustainable Tier 1 population. At 50% capacity, birth rate is halved. At 100%, births stop. |

### Condition (Health) System

| Field                                 | Type  | Default | Range    | Description                                                                                                                                                               |
|---------------------------------------|-------|---------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">ConditionDrainRate</a>    | float | 0.15    | 0.01-1.0 | How fast Condition drains toward poor performance. 0.15 = ~8 days from full health to death threshold at suboptimal temps. Drain accelerates up to 5x near lethal limits. |
| <a href="#">ConditionRecoveryRate</a> | float | 0.10    | 0.01-1.0 | How fast Condition recovers toward good performance. Intentionally slower than drain (asymmetric recovery). 0.10 = ~10 good days to fully recover.                        |

### Temperature: Base & Seasonal

| Field                             | Type  | Default | Description                                                                                   |
|-----------------------------------|-------|---------|-----------------------------------------------------------------------------------------------|
| <a href="#">BaseTemperature</a>   | float | 20      | Base/mean temperature in Celsius.                                                             |
| <a href="#">SeasonalAmplitude</a> | float | 10      | Amplitude of seasonal variation. Produces a sinusoidal summer/winter swing of +/- this value. |

### Temperature: Climate Trend

| Field                | Type  | Default | Description                                               |
|----------------------|-------|---------|-----------------------------------------------------------|
| ClimateTrend         | float | 1       | Degrees Celsius warming per year (linear climate change). |
| InterannualVariation | bool  | true    | Enable year-to-year random temperature variation.         |

### Temperature: Interannual Variation

| Field                | Type  | Default | Description                                                                                            |
|----------------------|-------|---------|--------------------------------------------------------------------------------------------------------|
| VariabilityMagnitude | float | 2       | Magnitude of year-to-year temperature variation. Each year gets a random offset drawn from this range. |
| WarmingBias          | float | 1.5     | Bias towards warmer years (positive skew). Higher = warm years more likely than cold years.            |

### Temperature: Daily Variation

| Field                | Type  | Default | Description                                                                                                                                                            |
|----------------------|-------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Autocorrelated       | bool  | true    | Enable autocorrelated (smooth) daily variation. When true, each day's random variation is 70% of yesterday's + 30% new random, creating realistic weather transitions. |
| DailyVariationRange  | float | 5       | Base daily random variation range in Celsius.                                                                                                                          |
| RandomnessGrowthRate | float | 0.5     | How much daily randomness increases per year. Models increasing climate instability.                                                                                   |

### Temperature: Bounds

| Field                | Type  | Default | Description                                           |
|----------------------|-------|---------|-------------------------------------------------------|
| TemperatureBoundsMin | float | -5      | Minimum possible temperature (hard floor, Celsius).   |
| TemperatureBoundsMax | float | 50      | Maximum possible temperature (hard ceiling, Celsius). |

### Species & Randomness

| Field      | Type           | Default                                                                                    | Description                                                                                                                                             |
|------------|----------------|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| RunSpecies | RunSpeciesList | Runtime species list ScriptableObject. Contains the actual species that will be simulated. |                                                                                                                                                         |
| RandomSeed | int            | 12345                                                                                      | Base random seed. -1 = use system time (non-reproducible). Any other value = reproducible results. Each scenario uses <b>BaseSeed + ScenarioIndex</b> . |

## Species Parameters

Species are defined in two parallel systems that share the same fields.

### SimSpecies (Simulation Runtime)

Source: [Assets/scripts/Simulation/SimSpecies.cs](#)

Used by the simulation engine during execution. Population stored as float for fractional precision.

### SpeciesData (Database/Inspector)

Source: [Assets/scripts/Simulation/DataStructure/SpeciesDatabase.cs](#)

Used for Unity Inspector editing and stored in ScriptableObjects. Mapped to **SimSpecies** at simulation start.

Note: `SpeciesData.tier` uses 0-indexed (0 = prey, 1 = predator), while `SimSpecies.Tier` uses 1-indexed (1 = prey, 2 = predator).

## Identity

| SimSpecies Field                 | SpeciesData Field                      | Type                                                                            | Description                                                                                 |
|----------------------------------|----------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Name                             | <code>displayName / speciesName</code> | string / enum                                                                   | Species name. SpeciesData has both an enum ( <code>SpeciesName</code> ) and display string. |
| Variant                          | <code>variant</code>                   | enum                                                                            | Thermal variant: Arctic, Common, Tropical, or Custom.                                       |
| Tier                             | <code>tier</code>                      | int                                                                             | Food chain level. SimSpecies: 1 = prey, 2 = predator. SpeciesData: 0 = prey, 1 = predator.  |
| Population                       | <code>count</code>                     | float / int                                                                     | Initial population. SimSpecies uses float for fractional tracking during simulation.        |
| (computed) <code>FullName</code> | string                                 | "{Name}_{Variant}" (e.g., "Hexapod_Arctic"). Used as dictionary key in results. |                                                                                             |

## Biological Parameters

| Field                               | Type  | T1 Default | T2 Default | Description                                                                                                                                                                                 |
|-------------------------------------|-------|------------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>EatingAmount</code>           | float | 0          | 1.5        | Prey consumed per creature per biology step. Tier 1 = 0 (they don't eat other species).                                                                                                     |
| <code>ReproductionMultiplier</code> | float | 0.45       | 0.1        | Birth rate multiplier. T2 reproduces 4.5x slower than T1.                                                                                                                                   |
| <code>DeathThreshold</code>         | float | 0.3        | 0.3        | Condition value below which condition deaths begin. Threshold-gated: no deaths above this.                                                                                                  |
| <code>DeathRate</code>              | float | 0.6        | 0.3        | Maximum fraction of population dying per day when below DeathThreshold. T1 higher due to allometric scaling (smaller prey have less physiological buffering).                               |
| <code>ReproThreshold</code>         | float | 0.25       | 0.25       | Condition inflection point for reproduction curve. Above: healthy ramp (0.10 to 1.0). Below: struggling but non-zero (0 to 0.10). Not a gate -- reproduction occurs whenever Condition > 0. |

## Natural Mortality

| Field                             | Type  | T1 Default  | T2 Default     | Description                                                                                                                                                                                                       |
|-----------------------------------|-------|-------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>NaturalDeathRate</code>     | float | 0.02 (2%)   | 0.01 (1%)      | Base natural death rate per biology step. Represents old age, disease, accidents. Always active, independent of performance. T2 lower due to allometric scaling (larger animals have lower background mortality). |
| <code>NaturalDeathVariance</code> | float | 0.01 (+-1%) | 0.005 (+-0.5%) | Random daily fluctuation around the base rate. Actual rate on any day = base +/- variance.                                                                                                                        |

## Hunting Efficiency (Tier 2 Only)

| Field                          | Type  | Default      | Description                                                                                                                     |
|--------------------------------|-------|--------------|---------------------------------------------------------------------------------------------------------------------------------|
| <code>HuntingEfficiency</code> | float | 0.75 (75%)   | Base hunting success rate at normal prey:predator ratio (20:1). Scaled by Holling Type II functional response. T1 ignores this. |
| <code>HuntingVariance</code>   | float | 0.15 (+-15%) | Random daily variance in hunting success.                                                                                       |

## Thermal Curve Parameters

All temperature parameters stored in Kelvin internally. Celsius values are computed for display (`K - 273.15`).

| Field         | Type  | Description                                                                             |
|---------------|-------|-----------------------------------------------------------------------------------------|
| OptimalTempK  | float | Temperature in Kelvin where the species performs best. The peak of the Arrhenius curve. |
| ArrhenBreadth | float | Arrhenius breadth parameter. Controls the width of the thermal performance curve.       |
| ArrhenLower   | float | Arrhenius lower parameter. Controls the left (cold) side drop-off rate.                 |
| ArrhenUpper   | float | Arrhenius upper parameter. Controls the right (warm) side drop-off rate.                |
| LowerBoundK   | float | Lower thermal bound in Kelvin. Inflection point for cold-side performance decline.      |
| UpperBoundK   | float | Upper thermal bound in Kelvin. Inflection point for warm-side performance decline.      |

### Peak Height & Lethal Limits

| Field             | Type  | Default                                                         | Description                                                                                                                                                      |
|-------------------|-------|-----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pmax              | float | 0.65 (Common), 0.85 (Arctic/Tropical), 1.0 (SimSpecies factory) | Maximum performance at optimal temperature (0-1). Scales the Arrhenius output. A Pmax of 0.65 means even at the perfect temperature, performance caps at 65%.    |
| CTminC            | float | -5 to 0 (varies)                                                | Critical thermal minimum in Celsius. At or below this temperature, performance = 0 (instant death). A smooth cosine fade of 2 degrees applies near the boundary. |
| CTmaxC            | float | 20 to 80 (varies)                                               | Critical thermal maximum in Celsius. At or above this temperature, performance = 0 (instant death). Same cosine fade applies.                                    |
| TemperatureDebuff | float | 0                                                               | Per-species temperature offset. Shifts the experienced temperature before thermal calculation. Positive = species "feels" warmer than ambient.                   |

### Runtime Values (Calculated Each Step)

These are set by **EcosystemSimulator** during the biology sequence. Not configurable.

| Field                 | Type  | Default                                                                                                     | Description                                                                                             |
|-----------------------|-------|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| RawThermalPerformance | float | Arrhenius curve output with CTmin/CTmax cosine fade applied, WITHOUT Pmax scaling. Range: 0-1.              |                                                                                                         |
| ThermalPerformance    | float | RawThermalPerformance * Pmax. Used for reproduction calculations.                                           |                                                                                                         |
| FedRate               | float | 1.0                                                                                                         | Feeding satisfaction (0-1). Tier 1 always 1.0. Tier 2 depends on hunting success and prey availability. |
| RawFinalPerformance   | float | RawThermalPerformance * FedRate. Target for Condition drain/recovery. Used for death checks (without Pmax). |                                                                                                         |
| FinalPerformance      | float | ThermalPerformance * FedRate. Used for reproduction threshold and birth calculations.                       |                                                                                                         |
| CurrentHuntingSuccess | float | This step's actual hunting success rate after Holling Type II + variance. For tracking/output only.         |                                                                                                         |

| Field     | Type  | Default | Description                                                                                                                                                                        |
|-----------|-------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Condition | float | 1.0     | Health/energy reserves [0-1]. Starts at 1.0. Drains toward RawFinalPerformance when environment is poor, recovers when good. Drives both reproduction scaling and condition death. |

## Temperature Model

Source: `Assets/scripts/Simulation/TemperatureCalculator.cs`

Formula:  $T(\text{day}) = \text{Base} + \text{Seasonal} + \text{Trend} + \text{InterannualVar} + \text{DailyVar}$ , then clamped to bounds.

| Field                | Type  | Default | Description                                                                                                                       |
|----------------------|-------|---------|-----------------------------------------------------------------------------------------------------------------------------------|
| BaseTemperature      | float | 20      | Starting temperature in Celsius.                                                                                                  |
| SeasonalAmplitude    | float | 10      | Seasonal swing via $\sin(2\pi \text{day}/365) * \text{amplitude}$ . Coldest at day 0, warmest at day 182.                         |
| ClimateTrendPerYear  | float | 1       | Linear warming: $\text{trend} * (\text{day} / 365)$ .                                                                             |
| VariabilityMagnitude | float | 2       | Year-to-year random offset range. Each year gets one value, constant for all 365 days.                                            |
| WarmingBias          | float | 1.5     | Skews interannual variation warm. Formula: $(\text{coldPart} + \text{warmPart} * \text{bias}) / 2$ .                              |
| BaseRandomness       | float | 5       | Daily random variation range in Celsius.                                                                                          |
| RandomnessGrowthRate | float | 0.5     | Daily randomness increases per year: $\text{currentRandomness} = \text{base} + \text{growth} * \text{yearNumber}$ .               |
| UseAutocorrelation   | bool  | true    | When true: $\text{variation} = 0.7 \text{ yesterday} + 0.3 \text{ newRandom}$ (smooth weather). When false: pure random each day. |
| MinTemp              | float | -5      | Hard floor -- temperature never goes below this.                                                                                  |
| MaxTemp              | float | 40      | Hard ceiling -- temperature never exceeds this.                                                                                   |

Constant: `DAYS_PER_YEAR` = 365

## Ecosystem Simulator

Source: `Assets/scripts/Simulation/EcosystemSimulator.cs`

### Configurable Properties

| Property                | Type  | Default | Description                                                   |
|-------------------------|-------|---------|---------------------------------------------------------------|
| BiologyStep             | int   | 1       | Days between biology calculations (mirrors SimulationConfig). |
| UseCarryingCapacity     | bool  | true    | Enable soft Tier 1 population limit.                          |
| CarryingCapacityPerTier | float | 5000    | Tier 1 population limit.                                      |
| ConditionDrainRate      | float | 0.15    | How fast Condition drains toward poor performance.            |
| ConditionRecoveryRate   | float | 0.10    | How fast Condition recovers toward good performance.          |

### Population Tracking (Read-Only, Per Step)

| Property   | Type  | Description                                      |
|------------|-------|--------------------------------------------------|
| StartPopT1 | float | Tier 1 population at start of this biology step. |
| StartPopT2 | float | Tier 2 population at start of this biology step. |

| Property | Type  | Description                                    |
|----------|-------|------------------------------------------------|
| EndPopT1 | float | Tier 1 population at end of this biology step. |
| EndPopT2 | float | Tier 2 population at end of this biology step. |

### Death/Birth Tracking (Read-Only, Per Step)

| Property                 | Type  | Description                                             |
|--------------------------|-------|---------------------------------------------------------|
| LastEatenT1              | float | Prey eaten this step (Tier 1 deaths from predation).    |
| LastTempDeathsT1         | float | Thermal deaths Tier 1 (instant kill at lethal limits).  |
| LastTempDeathsT2         | float | Thermal deaths Tier 2.                                  |
| LastConditionDeathsT1    | float | Condition deaths Tier 1 (chronic stress).               |
| LastConditionDeathsT2    | float | Condition deaths Tier 2.                                |
| LastNaturalDeathsT1      | float | Natural deaths Tier 1 (flat rate).                      |
| LastNaturalDeathsT2      | float | Natural deaths Tier 2.                                  |
| LastBirthsT1             | float | Births Tier 1 this step.                                |
| LastBirthsT2             | float | Births Tier 2 this step.                                |
| LastFedRateT2            | float | Tier 2 feeding satisfaction this step.                  |
| LastAvgHuntingEfficiency | float | Average hunting success rate this step.                 |
| LastReproScaleT1         | float | Condition-based reproduction scale factor for T1 [0-1]. |
| LastReproScaleT2         | float | Condition-based reproduction scale factor for T2 [0-1]. |

### Accumulator Totals (Read-Only, For CSV Output)

Accumulators carry fractional remainders between steps so no births or deaths are lost to rounding.

| Property              | Type  | Description                                      |
|-----------------------|-------|--------------------------------------------------|
| BirthAccumT1          | float | Birth accumulator residual for Tier 1.           |
| BirthAccumT2          | float | Birth accumulator residual for Tier 2.           |
| NaturalDeathAccumT1   | float | Natural death accumulator residual for Tier 1.   |
| NaturalDeathAccumT2   | float | Natural death accumulator residual for Tier 2.   |
| PredationAccumT1      | float | Predation accumulator residual for Tier 1.       |
| ConditionDeathAccumT1 | float | Condition death accumulator residual for Tier 1. |
| ConditionDeathAccumT2 | float | Condition death accumulator residual for Tier 2. |

### Condition Tracking

| Property       | Type  | Description                                       |
|----------------|-------|---------------------------------------------------|
| AvgConditionT1 | float | Population-weighted average Condition for Tier 1. |
| AvgConditionT2 | float | Population-weighted average Condition for Tier 2. |

### Biology Sequence (10 Steps)

Executed each biology step in this order:

- 1. Thermal Performance** - Arrhenius formula with CTmin/CTmax cosine fade
- 2. Feeding/Predation** - Holling Type II functional response + predation accumulator
- 3. Raw Final Performance** -  $\text{RawThermalPerf} * \text{FedRate}$  (Condition drain target)
- 4. Update Condition** - Drain or recover toward RawFinalPerformance

5. **Final Performance** - `ThermalPerf * FedRate` (Condition NOT used here)
6. **Thermal Death** - Instant kill when `RawThermalPerformance == 0`
7. **Condition Death** - Graduated severity when `Condition < DeathThreshold`, survivor fitness boost
8. **Reproduction** - Condition-based graduated scale + birth accumulator + carrying capacity
9. **Natural Death** - Flat rate + natural death accumulator
10. **Population Rounding** - All populations become integers (floor)

## Bulk CSV Import

Source: `Assets/scripts/Simulation/BulkBatchConfig.cs`

Each row in a bulk CSV file defines one run. These are the parsed column names.

### BulkBatchConfig (Per-Run Environment)

| Field                 | CSV Column              | Type   | Default                           | Description                                                     |
|-----------------------|-------------------------|--------|-----------------------------------|-----------------------------------------------------------------|
| BatchName             | batch_name              | string | Name/label for this run.          |                                                                 |
| Days                  | days                    | int    | Days per scenario.                |                                                                 |
| NumScenarios          | num_scenarios           | int    | Number of scenarios per run.      |                                                                 |
| BaseTemp              | base_temp               | float  | Base temperature (Celsius).       |                                                                 |
| SeasonalAmp           | seasonal_amp            | float  | Seasonal amplitude.               |                                                                 |
| ClimateTrend          | climate_trend           | float  | Climate warming per year.         |                                                                 |
| VariabilityMag        | variability_mag         | float  | Year-to-year variation magnitude. |                                                                 |
| WarmingBias           | warming_bias            | float  | Warm year bias.                   |                                                                 |
| DailyVarRange         | daily_var_range         | float  | Daily variation range.            |                                                                 |
| RandomnessGrowth      | randomness_growth       | float  | Daily randomness growth per year. |                                                                 |
| Autocorrelated        | autocorrelated          | bool   | Enable smooth daily transitions.  |                                                                 |
| InterannualVariation  | interannual_variation   | bool   | Enable year-to-year variation.    |                                                                 |
| TempMin               | temp_min                | float  | Temperature floor.                |                                                                 |
| TempMax               | temp_max                | float  | Temperature ceiling.              |                                                                 |
| UseCarryingCap        | use_carrying_cap        | bool   | Enable carrying capacity.         |                                                                 |
| CarryingCapT1         | carrying_cap_t1         | float  | Tier 1 carrying capacity.         |                                                                 |
| ConditionDrainRate    | condition_drain_rate    | float  | 0.15                              | Condition drain rate (optional column, backward compatible).    |
| ConditionRecoveryRate | condition_recovery_rate | float  | 0.10                              | Condition recovery rate (optional column, backward compatible). |

### BulkSpeciesConfig (Per-Species, Prefixed)

Species columns use a prefix pattern: `sp1_`, `sp2_`, `sp3_`, etc. All temperatures in Celsius.

| Field | CSV Suffix | Type   | Default                         | Description |
|-------|------------|--------|---------------------------------|-------------|
| Name  | name       | string | Species name (e.g., "Hexapod"). |             |

| Field            | CSV Suffix         | Type   | Default                                                                  | Description                         |
|------------------|--------------------|--------|--------------------------------------------------------------------------|-------------------------------------|
| Variant          | variant            | string | Variant name: "Arctic", "Common", "Tropical", or "Custom".               |                                     |
| Tier             | tier               | int    | 0 = Tier 1 (prey), 1 = Tier 2 (predator).                                |                                     |
| Pop              | pop                | int    | Initial population.                                                      |                                     |
| Eating           | eating             | float  | Prey consumed per creature per step.                                     |                                     |
| ReproMult        | repro_mult         | float  | Reproduction multiplier.                                                 |                                     |
| DeathThresh      | death_thresh       | float  | Condition death threshold.                                               |                                     |
| DeathRate        | death_rate         | float  | Condition death rate.                                                    |                                     |
| ReproThresh      | repro_thresh       | float  | Reproduction threshold (Condition inflection).                           |                                     |
| NaturalDeathRate | natural_death_rate | float  | Base natural death rate.                                                 |                                     |
| NaturalDeathVar  | natural_death_var  | float  | Natural death variance.                                                  |                                     |
| HuntEff          | hunt_eff           | float  | Base hunting efficiency.                                                 |                                     |
| HuntVar          | hunt_var           | float  | Hunting variance.                                                        |                                     |
| OptTempC         | opt_temp_c         | float  | Optimal temperature (Celsius). Converted to Kelvin (+273.15) internally. |                                     |
| ArrhenBreadth    | arrhen_breadth     | float  | Arrhenius breadth parameter.                                             |                                     |
| ArrhenLower      | arrhen_lower       | float  | Arrhenius lower parameter.                                               |                                     |
| ArrhenUpper      | arrhen_upper       | float  | Arrhenius upper parameter.                                               |                                     |
| LowerBoundC      | lower_bound_c      | float  | Lower thermal bound (Celsius). Converted to Kelvin internally.           |                                     |
| UpperBoundC      | upper_bound_c      | float  | Upper thermal bound (Celsius). Converted to Kelvin internally.           |                                     |
| Pmax             | pmax               | float  | 1.0                                                                      | Peak performance height.            |
| CTminC           | ctmin_c            | float  | -5.0                                                                     | Critical thermal minimum (Celsius). |
| CTmaxC           | ctmax_c            | float  | 50.0                                                                     | Critical thermal maximum (Celsius). |
| TempOffset       | temp_offset        | float  | 0                                                                        | Per-species temperature offset.     |

## Output: Scenario CSV

Source: `Assets/scripts/Simulation/SimulationRunner.cs` (`StepRecord` class)

Each scenario produces a CSV with `#config:` comment lines, a `#species:` table, then daily step data.

### Step Record Columns

| Column | Type | Description               |
|--------|------|---------------------------|
| Day    | int  | Simulation day (1-based). |



| Column                | Type  | Description                                                                  |
|-----------------------|-------|------------------------------------------------------------------------------|
| Year                  | int   | Year number (1-based, year = day/365 + 1).                                   |
| Temperature           | float | Temperature for this day (Celsius, F2 format).                               |
| BiologyCycle          | int   | Biology cycle counter. Increments only on days where biology runs.           |
| StartPop              | long  | Total population (T1+T2) at start of step.                                   |
| EndPop                | long  | Total population (T1+T2) at end of step.                                     |
| Tier1Pop              | long  | Tier 1 population at end of step.                                            |
| Tier2Pop              | long  | Tier 2 population at end of step.                                            |
| Tier1Arctic           | long  | Tier 1 Arctic variant population.                                            |
| Tier1Common           | long  | Tier 1 Common variant population.                                            |
| Tier1Tropical         | long  | Tier 1 Tropical variant population.                                          |
| Tier1Custom           | long  | Tier 1 Custom variant(s) population (sum of all custom T1 species).          |
| Tier2Arctic           | long  | Tier 2 Arctic variant population.                                            |
| Tier2Common           | long  | Tier 2 Common variant population.                                            |
| Tier2Tropical         | long  | Tier 2 Tropical variant population.                                          |
| Tier2Custom           | long  | Tier 2 Custom variant(s) population (sum of all custom T2 species).          |
| EatenT1               | long  | Prey eaten this step (T1 deaths from predation).                             |
| TempDeathsT1          | long  | Thermal deaths T1 (instant kill at lethal limits).                           |
| TempDeathsT2          | long  | Thermal deaths T2.                                                           |
| ConditionDeathsT1     | long  | Condition deaths T1 (chronic stress below threshold).                        |
| ConditionDeathsT2     | long  | Condition deaths T2.                                                         |
| NaturalDeathsT1       | long  | Natural deaths T1 (flat rate).                                               |
| NaturalDeathsT2       | long  | Natural deaths T2.                                                           |
| TotalDeaths           | long  | All deaths combined (predation + thermal + condition + natural, both tiers). |
| BirthsT1              | long  | New Tier 1 offspring this step.                                              |
| BirthsT2              | long  | New Tier 2 offspring this step.                                              |
| FedRateT2             | float | Tier 2 feeding satisfaction (F3 format, 0-1).                                |
| AvgHuntingEff         | float | Average hunting success rate (F3 format).                                    |
| AvgConditionT1        | float | Population-weighted average Condition for T1 (F3 format).                    |
| AvgConditionT2        | float | Population-weighted average Condition for T2 (F3 format).                    |
| BirthAccumT1          | float | Birth accumulator residual T1 (F3 format).                                   |
| BirthAccumT2          | float | Birth accumulator residual T2 (F3 format).                                   |
| NaturalDeathAccumT1   | float | Natural death accumulator residual T1 (F3 format).                           |
| NaturalDeathAccumT2   | float | Natural death accumulator residual T2 (F3 format).                           |
| ConditionDeathAccumT1 | float | Condition death accumulator residual T1 (F3 format).                         |
| ConditionDeathAccumT2 | float | Condition death accumulator residual T2 (F3 format).                         |
| PredationAccumT1      | float | Predation accumulator residual T1 (F3 format).                               |

| Column       | Type  | Description                                              |
|--------------|-------|----------------------------------------------------------|
| ReproScaleT1 | float | Condition-based reproduction scale T1 [0-1] (F3 format). |
| ReproScaleT2 | float | Condition-based reproduction scale T2 [0-1] (F3 format). |

## Config Comment Lines

Embedded as `#config:key,value` at the top of each scenario CSV. R's `read.csv()` ignores `#` lines.

| Key                     | Description                    |
|-------------------------|--------------------------------|
| days_per_scenario       | Days simulated                 |
| number_of_scenarios     | Total scenarios in this run    |
| scenario_index          | This scenario's 1-based index  |
| random_seed             | Seed used for this scenario    |
| biology_step            | Biology step interval          |
| use_carrying_capacity   | Carrying capacity enabled      |
| carrying_capacity_tier1 | T1 capacity limit              |
| condition_drain_rate    | Condition drain rate           |
| condition_recovery_rate | Condition recovery rate        |
| base_temperature        | Base temp (Celsius)            |
| seasonal_amplitude      | Seasonal swing                 |
| climate_trend           | Warming per year               |
| interannual_variation   | Year-to-year variation enabled |
| variability_magnitude   | Interannual variation range    |
| warming_bias            | Warm year bias                 |
| autocorrelated          | Smooth daily variation enabled |
| daily_variation_range   | Daily variation range          |
| randomness_growth_rate  | Randomness growth per year     |
| temperature_bounds_min  | Temp floor                     |
| temperature_bounds_max  | Temp ceiling                   |

## Output: ScenarioResult

Source: `Assets/scripts/Simulation/DataStructure/ScenarioResult.cs`

One per scenario. Contains summary stats and the full CSV string.

### Identity & Outcome

| Field         | Type | Description                                                                    |
|---------------|------|--------------------------------------------------------------------------------|
| ScenarioIndex | int  | 1-based index within the run.                                                  |
| RandomSeed    | int  | Seed used for this scenario.                                                   |
| TotalDays     | int  | Days simulated.                                                                |
| BiologyCycles | int  | Number of biology steps executed.                                              |
| Crashed       | bool | Did all populations go to zero? (Note: single-tier extinction is NOT a crash.) |
| CrashDay      | int  | Day of total extinction (-1 if no crash).                                      |
| CrashTier     | int  | Which tier triggered the crash check (-1 if no crash).                         |

## Final Populations

| Field                   | Type               | Description                                                                               |
|-------------------------|--------------------|-------------------------------------------------------------------------------------------|
| FinalTier1Pop           | long               | Final Tier 1 total population.                                                            |
| FinalTier2Pop           | long               | Final Tier 2 total population.                                                            |
| FinalTier1Arctic        | long               | Final T1 Arctic population.                                                               |
| FinalTier1Common        | long               | Final T1 Common population.                                                               |
| FinalTier1Tropical      | long               | Final T1 Tropical population.                                                             |
| FinalTier1Custom        | long               | Final T1 Custom population (sum of all custom T1).                                        |
| FinalTier2Arctic        | long               | Final T2 Arctic population.                                                               |
| FinalTier2Common        | long               | Final T2 Common population.                                                               |
| FinalTier2Tropical      | long               | Final T2 Tropical population.                                                             |
| FinalTier2Custom        | long               | Final T2 Custom population (sum of all custom T2).                                        |
| FinalSpeciesPopulations | Dict<string, long> | Per-species final populations keyed by FullName (e.g., "Hexapod_Arctic", "Coral_Custom"). |

## Population Extremes (Across All Days)

| Field       | Type | Description                                          |
|-------------|------|------------------------------------------------------|
| MaxTier1Pop | long | Maximum T1 population observed during this scenario. |
| MinTier1Pop | long | Minimum T1 population observed.                      |
| MaxTier2Pop | long | Maximum T2 population observed.                      |
| MinTier2Pop | long | Minimum T2 population observed.                      |

## Temperature Stats

| Field          | Type  | Description                          |
|----------------|-------|--------------------------------------|
| AvgTemperature | float | Average temperature across all days. |
| MinTemperature | float | Coldest day.                         |
| MaxTemperature | float | Warmest day.                         |

## Condition Stats

| Field            | Type  | Description                               |
|------------------|-------|-------------------------------------------|
| AvgConditionT1   | float | Average Condition for T1 across all days. |
| AvgConditionT2   | float | Average Condition for T2 across all days. |
| FinalConditionT1 | float | Condition T1 on the last day.             |
| FinalConditionT2 | float | Condition T2 on the last day.             |

## Per-Column Population Statistics

Keyed by column name (e.g., "Tier1Pop", "Tier1Arctic", etc.). See [PopColumns](#) and [VariantColumns](#) arrays.

| Field     | Type                 | Description                                 |
|-----------|----------------------|---------------------------------------------|
| PopMean   | Dict<string, double> | Mean population per column across all days. |
| PopMax    | Dict<string, long>   | Maximum population per column.              |
| PopMin    | Dict<string, long>   | Minimum population per column.              |
| PopStdDev | Dict<string, double> | Standard deviation per column.              |

| Field         | Type              | Description                                           |
|---------------|-------------------|-------------------------------------------------------|
| ExtinctionDay | Dict<string, int> | Day each variant first reached 0 (-1 if it survived). |

### Static Column Arrays

| Array          | Contents                                                                                                                                           |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| PopColumns     | "Tier1Pop", "Tier2Pop", "Tier1Arctic", "Tier1Common", "Tier1Tropical", "Tier1Custom", "Tier2Arctic", "Tier2Common", "Tier2Tropical", "Tier2Custom" |
| VariantColumns | "Tier1Arctic", "Tier1Common", "Tier1Tropical", "Tier1Custom", "Tier2Arctic", "Tier2Common", "Tier2Tropical", "Tier2Custom"                         |

## Output: AggregateResults

Source: `Assets/scripts/Simulation/DataStructure/ScenarioResult.cs`  
 Aggregates across all scenarios in one run. Computed by `CalculateAggregates()`.

### Run Info

| Field              | Type     | Description                                     |
|--------------------|----------|-------------------------------------------------|
| TotalScenarios     | int      | Total scenarios run.                            |
| CompletedScenarios | int      | Scenarios completed (currently always = Total). |
| SurvivedScenarios  | int      | Scenarios that did NOT crash.                   |
| CrashedScenarios   | int      | Scenarios that crashed (total extinction).      |
| CompletedAt        | DateTime | Timestamp when aggregation completed.           |

### Configuration (Stored for Export)

All `SimulationConfig` fields are copied here for config export:

`DaysPerScenario`, `BiologyStep`, `RandomSeed`, `UseCarryingCapacity`, `CarryingCapacity`, `ConditionDrainRate`, `ConditionRecoveryRate`, `BaseTemperature`, `SeasonalAmplitude`, `ClimateTrend`, `InterannualVariation`, `VariabilityMagnitude`, `WarmingBias`, `Autocorrelated`, `DailyVariationRange`, `RandomnessGrowthRate`, `TemperatureBoundsMin`, `TemperatureBoundsMax`, `RunSpecies`.

### Population Averages (Survived Scenarios Only)

| Field            | Type  | Description                                            |
|------------------|-------|--------------------------------------------------------|
| AvgFinalTier1Pop | float | Average final T1 population (survived scenarios only). |
| AvgFinalTier2Pop | float | Average final T2 population (survived scenarios only). |
| MinFinalTier1Pop | float | Minimum final T1 population across survived scenarios. |
| MaxFinalTier1Pop | float | Maximum final T1 population across survived scenarios. |
| MinFinalTier2Pop | float | Minimum final T2 population across survived scenarios. |
| MaxFinalTier2Pop | float | Maximum final T2 population across survived scenarios. |

### Crash Stats

| Field       | Type  | Description                                    |
|-------------|-------|------------------------------------------------|
| CrashRate   | float | Fraction of scenarios that crashed (0-1).      |
| AvgCrashDay | float | Average day of crash (crashed scenarios only). |

### Condition Stats

| Field               | Type  | Description                                                                          |
|---------------------|-------|--------------------------------------------------------------------------------------|
| AvgConditionT1      | float | Grand mean of per-scenario average Condition T1 (ALL scenarios, crashed + survived). |
| AvgConditionT2      | float | Grand mean of per-scenario average Condition T2 (ALL scenarios).                     |
| AvgFinalConditionT1 | float | Mean final Condition T1 (survived scenarios only).                                   |
| AvgFinalConditionT2 | float | Mean final Condition T2 (survived scenarios only).                                   |

### Per-Species Population Stats

Keyed by species `FullName` (e.g., "Hexapod\_Arctic", "Coral\_Custom"). Computed across ALL scenarios (including crashed).

| Field                 | Type                | Description                                                          |
|-----------------------|---------------------|----------------------------------------------------------------------|
| PerSpeciesAvg         | Dict<string, float> | Average final population per species (all scenarios).                |
| PerSpeciesMin         | Dict<string, float> | Minimum final population per species.                                |
| PerSpeciesMax         | Dict<string, float> | Maximum final population per species.                                |
| PerSpeciesExtinct     | Dict<string, int>   | Number of scenarios where species went extinct (final pop = 0).      |
| PerSpeciesSurvived    | Dict<string, int>   | Number of scenarios where species survived (final pop > 0).          |
| PerSpeciesSurvivedAvg | Dict<string, float> | Average population only across scenarios where the species survived. |

## Output: Bulk Summary

Source: `Assets/scripts/Simulation/BulkSimulationController.cs` (nested struct `BulkRunSummary`)

One per run (CSV row) in a bulk upload. Collected into `bulk_summary.csv` at the ZIP root.

| Field              | Type                | Description                                                        |
|--------------------|---------------------|--------------------------------------------------------------------|
| BatchName          | string              | Name/label from the CSV row.                                       |
| NumScenarios       | int                 | Total scenarios in this run.                                       |
| Survived           | int                 | Scenarios that survived.                                           |
| Crashed            | int                 | Scenarios that crashed.                                            |
| BaseTemp           | float               | Base temperature used.                                             |
| ClimateTrend       | float               | Climate trend used.                                                |
| AvgSpeciesPop      | Dict<string, float> | Average final population per species across all scenarios.         |
| SurvivedSpeciesPop | Dict<string, float> | Average final population per species only from survived scenarios. |

### Bulk Summary Aggregation Notes

- **GrandMean** = mean of per-run averages across all runs in the bulk
- **SurvivedMean** = mean of per-run survived averages across runs where the species had any presence

- **Extinction Rate** = per-species, computed from per-run data
- **Min/Max are NOT valid** at the bulk summary level (they would be min/max of averages, not real population values)

## Output: SimulationSummary

Source: `Assets/scripts/Simulation/SimulationRunner.cs`

Lightweight summary returned by `SimulationRunner.GetSummary()` after a single scenario completes.

| Field                           | Type  | Description                        |
|---------------------------------|-------|------------------------------------|
| <code>TotalDays</code>          | int   | Days simulated.                    |
| <code>TotalBiologyCycles</code> | int   | Biology cycles executed.           |
| <code>Crashed</code>            | bool  | Did total extinction occur?        |
| <code>CrashDay</code>           | int   | Day of crash.                      |
| <code>CrashTier</code>          | int   | Tier that triggered crash.         |
| <code>FinalTier1Pop</code>      | long  | Final T1 population.               |
| <code>FinalTier2Pop</code>      | long  | Final T2 population.               |
| <code>FinalTier1Arctic</code>   | long  | Final T1 Arctic.                   |
| <code>FinalTier1Common</code>   | long  | Final T1 Common.                   |
| <code>FinalTier1Tropical</code> | long  | Final T1 Tropical.                 |
| <code>FinalTier1Custom</code>   | long  | Final T1 Custom.                   |
| <code>FinalTier2Arctic</code>   | long  | Final T2 Arctic.                   |
| <code>FinalTier2Common</code>   | long  | Final T2 Common.                   |
| <code>FinalTier2Tropical</code> | long  | Final T2 Tropical.                 |
| <code>FinalTier2Custom</code>   | long  | Final T2 Custom.                   |
| <code>MaxTier1Pop</code>        | long  | Max T1 population during scenario. |
| <code>MinTier1Pop</code>        | long  | Min T1 population during scenario. |
| <code>MaxTier2Pop</code>        | long  | Max T2 population during scenario. |
| <code>MinTier2Pop</code>        | long  | Min T2 population during scenario. |
| <code>AvgTemperature</code>     | float | Average temperature.               |
| <code>MinTemperature</code>     | float | Minimum temperature.               |
| <code>MaxTemperature</code>     | float | Maximum temperature.               |

## Enums

### *ThermalVariant*

Source: `Assets/scripts/Simulation/SimSpecies.cs`

Used by the simulation runtime.

| Value                 | Description                                                              |
|-----------------------|--------------------------------------------------------------------------|
| <code>Arctic</code>   | Cold-adapted species.                                                    |
| <code>Common</code>   | Temperate/generalist species.                                            |
| <code>Tropical</code> | Warm-adapted species.                                                    |
| <code>Custom</code>   | User-defined species from bulk CSV import. Tracked individually by name. |

### *SpeciesName*

Source: `Assets/scripts/Simulation/DataStructure/SpeciesDatabase.cs`

Used by the game/inspector system.

| Value    | Tier         | Description                                    |
|----------|--------------|------------------------------------------------|
| Hexapod  | 1 (prey)     | Primary Tier 1 species.                        |
| Gelgi    | 1            | Tier 1 species (game only).                    |
| Yelloa   | 1            | Tier 1 species (game only).                    |
| Sheplik  | 2 (predator) | Primary Tier 2 species.                        |
| Grabbler | 2            | Tier 2 species (game only).                    |
| Cyplo    | 2            | Tier 2 species (game only).                    |
| Rooda    | 3            | Tier 3 species (game only, not in simulation). |
| Sploof   | 3            | Tier 3 species (game only).                    |
| Silu     | 3            | Tier 3 species (game only).                    |
| Custom   | any          | User-defined species.                          |

## SpeciesVariant

Source: `Assets/scripts/Simulation/DataStructure/SpeciesDatabase.cs`

Used by the game/inspector system. Note different ordering from `ThermalVariant`.

| Value    | Description           |
|----------|-----------------------|
| Common   | Temperate/generalist. |
| Tropical | Warm-adapted.         |
| Arctic   | Cold-adapted.         |
| Custom   | User-defined.         |

## Constants

All hardcoded constants across the simulation system.

### EcosystemSimulator Constants

| Constant                        | Value | Description                                                                                                                      |
|---------------------------------|-------|----------------------------------------------------------------------------------------------------------------------------------|
| MIN_ALIVE_POP                   | 1.0   | Minimum population to be considered alive.                                                                                       |
| DRAIN_ACCEL_THRESHOLD           | 0.2   | Performance below this accelerates Condition drain.                                                                              |
| DRAIN_ACCEL_MAX                 | 4.0   | Maximum drain acceleration multiplier (total 5x at perf=0, because it's $1 + severity * 4$ ).                                    |
| NEWBORN_CONDITION               | 0.5   | Condition value for newborn individuals. Dilutes group average.                                                                  |
| NORMAL_PREY_RATIO               | 20.0  | Prey:predator ratio where base hunting efficiency applies (Holling Type II half-saturation reference).                           |
| MIN_HUNTING_SUCCESS             | 0.0   | Floor for hunting success. Zero prey = zero hunting.                                                                             |
| MAX_HUNTING_SUCCESS             | 1.0   | Ceiling for hunting success.                                                                                                     |
| MIN_POPULATION_FOR_REPRODUCTION | 2.0   | Minimum population required to reproduce (need at least 2).                                                                      |
| STRUGGLING_REPRO_RATE           | 0.10  | Maximum reproScale when Condition equals ReproThreshold. The two piecewise regions of the reproduction curve meet at this value. |

### SimSpecies Constants

| Constant                         | Value | Description                                                                              |
|----------------------------------|-------|------------------------------------------------------------------------------------------|
| NO_PREDATOR_PENALTY              | 0.85  | T1 birth multiplier when no predators exist (15% reduction to prevent unchecked growth). |
| MIN_FINAL_PERF_FOR_NATURAL_DEATH | 0.1   | Floor for FinalPerformance in natural death calculation to prevent division by zero.     |
| LETHAL_TRANSITION_WIDTH          | 2.0   | Width in Celsius of the smooth cosine fade at CTmin/CTmax boundaries.                    |

TemperatureCalculator Constants

| Constant      | Value | Description                              |
|---------------|-------|------------------------------------------|
| DAYS_PER_YEAR | 365   | Days per year for seasonal calculations. |

SpeciesDatabase Constants

| Constant         | Value | Description                                    |
|------------------|-------|------------------------------------------------|
| DEFAULT_T1_COUNT | 20    | Default initial Tier 1 population.             |
| DEFAULT_T2_COUNT | 4     | Default initial Tier 2 population.             |
| DEFAULT_T3_COUNT | 2     | Default initial Tier 3 population (game only). |

Default Species Values (Quick Reference)

| Parameter              | Hexapod (T1) | Sheplik (T2) |
|------------------------|--------------|--------------|
| EatingAmount           | 0            | 1.5          |
| ReproductionMultiplier | 0.45         | 0.1          |
| DeathThreshold         | 0.3          | 0.3          |
| DeathRate              | 0.6          | 0.3          |
| ReproThreshold         | 0.25         | 0.25         |
| NaturalDeathRate       | 2%           | 1%           |
| NaturalDeathVariance   | +1%          | +0.5%        |
| HuntingEfficiency      | N/A          | 75%          |
| HuntingVariance        | N/A          | +15%         |

Default Thermal Variants (from SpeciesDatabase)

| Variant  | OptimalTemp (C) | Arrhen Breadth | Arrhen Lower | Arrhen Upper | Lower Bound (C) | Upper Bound (C) | Pmax | CTmin (C) | CTmax (C) |
|----------|-----------------|----------------|--------------|--------------|-----------------|-----------------|------|-----------|-----------|
| Common   | 24              | 8000           | 3000         | 35000        | 23              | 25              | 0.65 | 0         | 40        |
| Tropical | 30              | 4000           | 15827        | 35000        | 29.75           | 29.95           | 0.85 | 0         | 40        |
| Arctic   | 18              | 4000           | 13974        | 35000        | 17.75           | 17.95           | 0.85 | 0         | 40        |